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THE TRUNCATED NEGATIVE BINOMIAL DISTRIBUTION

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1. INTRODUCTION

The negative binomial distribution has been discussed by, *inter alia*, Greenwood & Yule (1920), Fisher (1941), Haldane (1941), Anscombe (1950) and Bliss & Fisher (1953), and is extensively used for the description of data too heterogeneous to be fitted by a Poisson distribution. Observed samples, however, may be truncated, in the sense that the number of individuals falling into the zero class cannot be determined. For example, if chromosome breaks in irradiated tissue can occur only in those cells which are at a particular stage of the mitotic cycle at the time of irradiation, a cell can be demonstrated to have been at that stage only if breaks actually occur. Thus in the distribution of breaks per cell, cells not susceptible to breakage are indistinguishable from susceptible cells in which no breaks occur.

Methods for estimation of the parameters of the truncated distribution are considered in this paper. The corresponding problem of estimation of the truncated Poisson distribution has been discussed by David & Johnson (1952), who also discuss the present problem.

2. THE MOMENTS OF THE TRUNCATED DISTRIBUTION

The negative binomial distribution has the form (in Fisher's notation)

$$P(r) = \frac{(k+r-1)!}{(k-1)! r!} \frac{p^r}{(1+p)^{k+r}} \quad (r = 0, 1, \dots; p, k > 0), \quad (1)$$

so that

$$P(0) = 1/(1+p)^k.$$

To obtain the corresponding probabilities for the truncated distribution the form (1) must be divided by $(1 - P(0))$; writing

$$\varpi = 1/(1+p), \quad \eta = 1 - \varpi,$$

it follows that

$$P_t(r) = \frac{\varpi^k}{1 - \varpi^k} \frac{(k+r-1)!}{(k-1)! r!} \eta^r \quad (r = 1, 2, \dots). \quad (2)$$

The factorial moments of this distribution are

$$\mu'_{[j]} = \frac{(k+j-1)! \eta^j}{(k-1)! \varpi^j (1 - \varpi^k)}, \quad (3)$$

whence

$$\begin{aligned} \mu'_1 &= \frac{k\eta}{\varpi(1 - \varpi^k)}, \\ \mu'_2 &= \frac{k\eta + k^2\eta^2}{\varpi^2(1 - \varpi^k)}, \\ \mu'_3 &= \frac{k\eta + k(3k+1)\eta^2 + k^3\eta^3}{\varpi^3(1 - \varpi^k)}, \\ \mu'_4 &= \frac{k\eta(6 - 6\varpi + \varpi^2) + k^2\eta^2(11 - 4\varpi) + 6k^3\eta^3 + k^4\eta^4}{\varpi^4(1 - \varpi^k)}. \end{aligned}$$

3. ESTIMATION BY MOMENTS

David & Johnson (1952) suggest that the use of estimates of less than maximum efficiency is justifiable only if they are directly obtainable as explicit solutions of easily constructed equations. In discussing the truncated negative binomial, therefore, they do not consider estimates based on the first two sample moments, which do not provide explicit solutions, but confine their attention to a method using certain ratios of the first three moments, i.e. sample estimates of μ'_2/μ'_1 and μ'_3/μ'_1 . The estimates (only that of p is discussed in detail) are obtained easily enough, but, in consequence of the introduction of the third sample moment, are extremely inefficient. (For example, the efficiency for values of the parameters equivalent to $k = 1$, $\varpi = 0.5$ is as low as 1.7 %.) David & Johnson therefore abandon completely the use of moments and recommend the maximum-likelihood method for use in all cases.

Whether, in fact, any particular inefficient procedure is acceptable can only depend on the loss of information resulting from its use, the time saved by it, and the relative costs of the time or labour spent on observation and on analysis. Thus if an experiment involves observations on hundreds of experimental animals, made over a period of several years, ten or even a hundred hours of calculation may be dearly saved at the cost of 10 % of the information so laboriously accumulated. If, on the other hand, the observations are made easily and at no great cost, the use of a convenient but statistically 'inefficient' method of analysis, coupled with an appropriate increase in sample size, may be far more 'efficient' than a tedious maximum-likelihood calculation, in the sense of giving the same amount of information at a lower cost. In this section I give a trial-and-error method for solving the moment equations using the first two moments. This method, though not explicit, does not take many minutes to carry through. It certainly entails far less labour than the maximum-likelihood calculations, and the estimates obtained have a percentage efficiency of 80 or upwards for all but the most unfavourable combinations of parameters: for $k = 1$, $\varpi = 0.5$ the efficiency is 77.5 % (§ 6, Table 2), as compared with the 1.7 % of the 'three-moment' estimate. This method can thus be recommended as a reasonable alternative to the maximum-likelihood calculations, in circumstances where a method of less than 100 % efficiency seems likely to prove acceptable.

Equating the population mean and variance to the corresponding sample values gives equations for k and ϖ :

$$\left. \begin{aligned} \frac{k\eta}{\varpi(1-\varpi^k)} &= m, \\ \frac{k\eta(1+k\eta)}{\varpi^2(1-\varpi^k)} &= m^2 + s^2. \end{aligned} \right\} \quad (4)$$

These equations can be solved by trial and error, for which purpose they are most conveniently expressed in the form

$$\left. \begin{aligned} k &= \left(m + \frac{s^2}{m} - 1\right) \frac{\varpi}{1-\varpi} - 1, \\ \xi &= m\varpi^{k+1} + \frac{s^2}{m}\varpi - 1 = 0. \end{aligned} \right\} \quad (5)$$

The value of ξ can be evaluated for selected values of ϖ , and a solution, if one exists, reached by successive linear interpolations.

To investigate whether a solution exists, and to simplify still further the computations required, we consider the function

$$\phi(x) = \frac{-x \log_e x}{1-x}. \quad (6)$$

This function is tabulated, for values of x between 0 and 1, in Table 4, and can be shown to possess the properties

$$\begin{aligned} \phi(x) &\rightarrow 0, 1 \quad \text{as } x \rightarrow 0, 1, \\ \phi'(x) &> 0, \quad \phi''(x) < 0, \quad 0 < x < 1. \end{aligned}$$

The second of equations (5) then takes the form

$$\zeta(\varpi) = m \exp \left\{ - \left(m + \frac{s^2}{m} - 1 \right) \phi(\varpi) \right\} + \frac{s^2}{m} \varpi = 1. \quad (7)$$

If all the observed values equal 1, $m = 1$ and $s^2 = 0$, and $\zeta \equiv 1$. Thus in this situation the moment estimation procedure, naturally enough, fails. In all further discussion, therefore, m will be assumed > 1 .

It is easily verified that

$$\varpi_0 = 1 / \left(m + \frac{s^2}{m} \right) \quad (< 1)$$

satisfies equation (7). However, the corresponding value of k , obtained from the first of equations (5), is 0. ϖ_0 , therefore, and any lower values of ϖ (corresponding to negative values of k) are inadmissible: we require a solution of (7) in the range

$$\varpi_0 < \varpi < 1.$$

We have $\zeta(0) = m > 1$, $\zeta(\varpi_0) = 1$, $\zeta'' > 0$ ($0 < \varpi < 1$).

(The result on ζ'' follows from that on ϕ'' quoted above.) There exists, therefore, at most one solution of (7) other than ϖ_0 in the range $0 < \varpi < 1$. The condition that there shall be another solution less than 1 is

$$\zeta(1) > 1,$$

the condition that it shall be greater than ϖ_0 is

$$\zeta'(\varpi_0) < 0.$$

These inequalities reduce to limitations on the relative magnitudes of m and s^2 , most conveniently expressed in the form

$$1 - \exp \left\{ - \left(m + \frac{s^2}{m} - 1 \right) \right\} < \left(m + \frac{s^2}{m} - 1 \right) / m < \log \left(m + \frac{s^2}{m} \right). \quad (8)$$

Neither of these inequalities is necessarily satisfied; it is quite possible to construct samples in which they are not. In particular, the lower inequality is not satisfied if $s^2 = 0$, whatever the value of m (reasonably enough; we should hardly expect to get sensible estimates by equating an essentially positive function to zero). However, provided very small samples are avoided, and provided m is not too near 1, (8) will probably be satisfied.

Equation (7) can be solved either iteratively, writing

$$\varpi_{i+1} = \frac{m}{s^2} \left\{ 1 - m \exp \left[- \left(m + \frac{s^2}{m} - 1 \right) \phi(\varpi_i) \right] \right\}, \quad (9)$$

or by trial and error. Inasmuch as the solution must lie on that part of the ζ curve for which $\zeta' > 0$, between the (unknown) minimum of ζ and the point $\varpi = 1$, it is most convenient to take as the first trial value of ϖ either 1, or, if it is less than 1, $\varpi = m/s^2$ (clearly $\zeta > 1$ for this value), and to work down to the solution. This approach will ensure the convergence of the iterative procedure (9), and should minimize the amount of labour required for the trial-and-error procedure. In the latter case time will probably be saved by calculating, for the first trial value of ϖ , the slope

$$\zeta'(\varpi) = \frac{s^2}{m} - m \left(m + \frac{s^2}{m} - 1 \right) \left[\frac{\phi(\varpi) - \varpi}{\varpi(1 - \varpi)} \right] \exp \left\{ - \left(m + \frac{s^2}{m} - 1 \right) \phi(\varpi) \right\}. \quad (10)$$

Except for the function in square brackets ($= \phi'(\varpi)$) all the terms used in calculating this expression will already have been evaluated in the calculation of ζ . This slope can be used in choosing the second trial value, after which one or two linear interpolations or extrapolations should be sufficient. If $\varpi = 1$ is taken as the initial approximate, $\phi'(\varpi)$ should be replaced by its limiting value of $\frac{1}{2}$.

4. EXAMPLE OF ESTIMATION BY MOMENTS

In an investigation into chromosome breakage, the following sample distribution of breaks per cell was obtained:

$$r = 1(11), 2(6), 3(4), 4(5), 6(1), 8(2), 9, 11, 13.$$

$$n = 32, \quad \Sigma r = 110, \quad \Sigma r^2 = 686.$$

$$m = 3.4375, \quad s^2 = 9.9315, \quad s^2/m = 2.8892.$$

Equation (7) is thus

$$3.4375 e^{-5.3267\phi} + 2.8892\varpi = 1.$$

Taking as the first trial value

$$\varpi = 1/2.8892 = 0.3461,$$

we obtain from Table 4

$$\phi = 0.5616,$$

whence

$$(\phi - \varpi)/\varpi(1 - \varpi) = 0.9522.$$

Then

$$5.3267\phi = 2.9915,$$

and from tables of the negative exponential function, or of natural logarithms,

$$e^{-2.9915} = 0.05021, \quad 3.4375 e^{-2.9915} = 0.1726,$$

whence

$$\zeta = 0.1726 + 1.0000 = 1.1726$$

and

$$\zeta' = 2.8892 - (0.1726 \times 5.3267 \times 0.9522) = 2.0138,$$

suggesting, as the next trial value,

$$\varpi = 0.3461 - 0.1726/2.0138 = 0.2604.$$

The remainder of the trial-and-error solution, shown in Table 1 (a), leads to the value

$$\varpi = 0.2346,$$

whence

$$k = 5.3267 \times 0.2346 / 0.7654 - 1 = 0.633.$$

Alternatively, by the iterative method, starting from the same initial value, the second approximation

$$\varpi_2 = (1 - 0.1726) / 2.8892 = 0.2864.$$

The remainder of the calculations, leading after sixteen cycles to the same solution, are shown in Table 1 (b).

Table 1. *Solution of the moment estimation equation for the example of § 4*

(a) By trial and error

ϖ	ϕ	$3.4375 e^{-5.3267\phi}$	ζ	ζ'
0.3461	0.5616	0.1726	1.1726	2.0137
0.2604	0.4737	0.2757	1.0280	
0.2438	0.4550	0.3046	1.0090	
0.2359	0.4459	0.3197	1.0013	
0.2346	0.4444	0.3222	1.0000	

(b) By iteration

ϖ	ϕ	$3.4375 e^{-5.3267\phi}$
0.3461	0.5616	0.1726
0.2864	0.5018	0.2374
0.2639	0.4776	0.2700
0.2527	0.4651	0.2886
0.2462	0.4578	0.3000
0.2423	0.4533	0.3073
0.2397	0.4503	0.3123
0.2380	0.4484	0.3154
0.2370	0.4472	0.3175
0.2362	0.4463	0.3190
0.2357	0.4457	0.3200
0.2353	0.4452	0.3209
0.2350	0.4448	0.3216
0.2348	0.4446	0.3219
0.2347	0.4445	0.3221
0.2346	0.4444	0.3222
0.2346		

(Intermediate stages of the calculations are shown in Table 1 for the sake of the example. In practice either calculation can be carried through on a desk computer, recording only the successive values of ϖ and, for the trial-and-error process, ζ .)

Four decimal places have been retained in the estimation of ϖ as a demonstration of the degree of precision to which the moment equations can be solved by this method. In fact, for this example the variances (§ 5) are so large that there is no practical point in retaining more than two decimal places for ϖ and one for k .

5. THE VARIANCES OF THE MOMENT ESTIMATES

The asymptotic variances and covariances of the moment estimates of ϖ and k are given by the formulae

$$\begin{aligned} J^2 V(\varpi) &= \left(\frac{\partial \mu'_2}{\partial k}\right)^2 V(m'_1) - 2 \frac{\partial \mu'_1}{\partial k} \frac{\partial \mu'_2}{\partial k} \text{cov}(m'_1, m'_2) + \left(\frac{\partial \mu'_1}{\partial k}\right)^2 V(m'_2), \\ J^2 \text{cov}(\varpi, k) &= -\frac{\partial \mu'_2}{\partial k} \frac{\partial \mu'_2}{\partial \varpi} V(m'_1) + \left(\frac{\partial \mu'_1}{\partial k} \frac{\partial \mu'_2}{\partial \varpi} + \frac{\partial \mu'_1}{\partial \varpi} \frac{\partial \mu'_2}{\partial k}\right) \text{cov}(m'_1, m'_2) - \frac{\partial \mu'_1}{\partial k} \frac{\partial \mu'_1}{\partial \varpi} V(m'_2), \\ J^2 V(k) &= \left(\frac{\partial \mu'_2}{\partial \varpi}\right)^2 V(m'_1) - 2 \frac{\partial \mu'_1}{\partial \varpi} \frac{\partial \mu'_2}{\partial \varpi} \text{cov}(m'_1, m'_2) + \left(\frac{\partial \mu'_1}{\partial \varpi}\right)^2 V(m'_2), \end{aligned}$$

where

$$\begin{aligned} \frac{\partial \mu'_1}{\partial k} &= \frac{\mu'_1}{k} \{1 - \phi(\varpi^k)\}, \\ \frac{\partial \mu'_1}{\partial \varpi} &= -\frac{\mu'_1}{\varpi \eta} \{1 - \varpi^{k+1} \mu'_1\}, \\ \frac{\partial \mu'_2}{\partial k} &= -\frac{\mu'_1}{k \varpi} + \frac{\mu'_2}{k} \{2 - \phi(\varpi^k)\}, \\ \frac{\partial \mu'_2}{\partial \varpi} &= \frac{\mu'_1}{\varpi \eta} - \frac{\mu'_2}{\varpi \eta} \{2 - \varpi^{k+1} \mu'_1\}, \\ J &= \frac{\partial \mu'_1}{\partial k} \frac{\partial \mu'_2}{\partial \varpi} - \frac{\partial \mu'_1}{\partial \varpi} \frac{\partial \mu'_2}{\partial k}, \end{aligned}$$

and $n V(m'_1) = (\mu'_2 - \mu'^2_1)$, $n \text{cov}(m'_1, m'_2) = (\mu'_3 - \mu'_1 \mu'_2)$, $n V(m'_2) = (\mu'_4 - \mu'^2_2)$.

Inserting into these formulae the moment estimates obtained in the example of the previous section,

$$\varpi = 0.2346, \quad k = 0.633,$$

we have ($\log_e \varpi = -1.44988$, $\varpi^k = 0.3994$):

$$\mu'_1 = 3.438578, \quad \mu'_2 = 21.758578, \quad \mu'_3 = 215.773952, \quad \mu'_4 = 2941.290671,$$

whence $V(m'_1) = 0.31046123$, $\text{cov}(m'_1, m'_2) = 4.4048558$, $V(m'_2) = 77.120467$.

Also $\partial \mu'_1 / \partial k = 2.1164$, $\partial \mu'_1 / \partial \varpi = -12.9798$, $\partial \mu'_2 / \partial k = 24.6106$, $\partial \mu'_2 / \partial \varpi = -184.1591$

$$J = -70.31365336,$$

whence $V(\varpi) = 0.015091$, $\text{cov}(\varpi, k) = 0.08125$, $V(k) = 0.4983$.

6. THE EFFICIENCY OF THE MOMENT PROCEDURE

It has been shown by Fisher (1941) that the method of moments may be seriously inefficient for the estimation, from complete data, of the parameters of the negative binomial distribution. Investigation of the corresponding efficiency for estimation from truncated data therefore seems desirable.

The determinant of the variance-covariance matrix of the moment estimates reduces to

$$\frac{2(k+1) \varpi^2 (1 - \varpi^k)^3}{n^2 \eta} \times \frac{\{1 - \varpi^k [1 + k\eta + \frac{1}{2}k(k+1)\eta^2]\}}{\{1 - \varpi^k [1 - k^2\eta - k(k+1)\log \varpi]\}^2}.$$

The determinant of the information matrix for maximum efficiency is

$$\frac{n^2 \{1 - \varpi^k [1 + k\eta]\}}{\varpi^2 \eta (1 - \varpi^k)^3} \left\{ \sum_{r=2}^{\infty} \frac{\eta^r}{r} \frac{(r-1)! k!}{(k+r-1)!} - \frac{k \varpi^k (\eta + \log \varpi)^2}{\{1 - \varpi^k [1 + k\eta]\}} \right\},$$

whence the reciprocal of the efficiency of the moment method, given by the product of these two expressions, is

$$\frac{1}{E} = \frac{\{1 - \varpi^k[1 + k\eta + \frac{1}{2}k(k+1)\eta^2]\} \{1 - \varpi^k[1 + k\eta]\}}{\{1 - \varpi^k[1 - k^2\eta - k(k+1)\log \varpi]\}^2} \times \left\{ 2 \sum_{r=2}^{\infty} \frac{\eta^{r-2}}{r} \frac{(r-1)!(k+1)!}{(k+r-1)!} - \frac{2k(k+1)\varpi^k}{[1 - \varpi^k(1 + k\eta)]} \left[1 + \frac{\log \varpi}{\eta} \right]^2 \right\}.$$

Table 2 shows the percentage efficiency of the moment method for selected values of k , and of the mean of the complete distribution. From this it is clear that even for quite small means there may be a serious loss of efficiency for low values of k .

For the example of § 4, k was estimated as 0.633, and the mean of the complete distribution = $k\eta/\varpi = 2.065$, from which estimation by moments would appear to be about 70 % efficient in this case.

Table 2. *Percentage efficiency of the moment method of estimation, for selected values of k and of the mean of the complete distribution*

k Mean	0.5	1	2	3	4	5
0.5	82.4	90.6	95.7	97.5	98.3	98.8
1.0	74.6	84.7	92.4	95.4	96.9	97.8
2.0	66.1	77.5	87.5	92.0	94.4	95.9
5.0	55.9	67.3	79.2	85.5	89.2	91.6

7. ESTIMATION BY MAXIMUM LIKELIHOOD

In view of the results of the previous section, it seems desirable to consider the maximum-likelihood estimation procedure in some detail.

The log likelihood is

$$\begin{aligned} \log L &= \sum_1^{\infty} n_r \log \left\{ \frac{\varpi^k}{1 - \varpi^k} \frac{(k+r-1)!}{(k-1)! r!} (1-\varpi)^r \right\} \\ &= nk \log \varpi - n \log (1 - \varpi^k) + \sum_1^{\infty} r n_r \log (1 - \varpi) - \sum_1^{\infty} n_r \log r! + \sum_{r=1}^{\infty} n_r \sum_{j=1}^r \log (k+j-1), \end{aligned}$$

giving the maximum-likelihood equations

$$\frac{nk}{\varpi(1 - \varpi^k)} - \frac{nm}{(1 - \varpi)} = 0 \quad (11)$$

and

$$\frac{n \log \varpi}{1 - \varpi^k} + \sum_{r=1}^{\infty} n_r \sum_{j=1}^r \frac{1}{(k+j-1)} = 0, \quad (12)$$

which, following Haldane (1941), is conveniently rewritten as

$$\frac{n \log \varpi}{1 - \varpi^k} + \sum_{j=1}^R (k+j-1)^{-1} \cdot \sum_{i=j}^R n_i = 0, \quad (13)$$

where R is the highest observed value of r .

These equations are easily soluble by the usual maximum-likelihood iterative procedure. The components of the information matrix are the expected values of the quantities

$$\left. \begin{aligned} -\frac{\partial^2 \log L}{\partial \varpi^2} &= \frac{nk[1 - (k+1)\varpi^k]}{\varpi^2(1-\varpi^k)^2} + \frac{nm}{(1-\varpi)^2}, \\ -\frac{\partial^2 \log L}{\partial \varpi \partial k} &= -\frac{n[1 - (1-k \log \varpi)\varpi^k]}{\varpi(1-\varpi^k)^2}, \\ -\frac{\partial^2 \log L}{\partial k^2} &= \sum_{j=1}^R (k+j-1)^{-2} \sum_{i=j}^R n_i - \frac{n(\log \varpi)^2 \varpi^k}{(1-\varpi^k)^2}. \end{aligned} \right\} \quad (14)$$

The iteration, however, is most conveniently carried out using the quantities (14) themselves, rather than their expected values. An example of the calculations for this method is given at the end of the next section.

8. AN ALTERNATIVE METHOD FOR THE SOLUTION OF THE MAXIMUM-LIKELIHOOD EQUATIONS

From equations (11) and (13) it follows that

$$\phi(\hat{\varpi}) = -\frac{\hat{\varpi} \log_e \hat{\varpi}}{1 - \hat{\varpi}} = \frac{\hat{k}}{nm} \sum_{j=1}^R (\hat{k} + j - 1)^{-1} \sum_{r=j}^R n_r, \quad (15)$$

where $\hat{\varpi}$ and $\hat{k}$ are maximum-likelihood estimates, and m is the sample mean. Equation (11) can be rewritten

$$\psi(\hat{\varpi}, \hat{k}) = \frac{\hat{\varpi}(1 - \hat{\varpi}^k)m}{\hat{k}(1 - \hat{\varpi})} = 1. \quad (16)$$

For a given value of k , equation (15) can be used to evaluate the corresponding ϕ and hence, from Table 4, ϖ ; the value of ψ can then be calculated. This form of the equations was presented by David & Johnson (1952); they suggested an iterative method, equivalent in the present notation to

$$k_{i+1} = [k_i \psi(\varpi_i, k_i)],$$

and provided a table of values of $p = (1 - \varpi)/\varpi$ for values of ϕ (in their notation, y) 0.40 (0.05) 1.00. This process, however, appears to converge rather slowly, and a solution can, in many cases, be reached more expeditiously by a trial-and-error process, starting with the moment, or some other inefficient, estimate of k . For this value of k the value of ψ can be evaluated, together with the slope of the curve ($\psi = \psi(k)$) defined by (15) and (16):

$$\psi'(k) = \frac{1}{\phi - \varpi} [\psi - m\varpi^{k+1}] \left[\frac{\phi}{k} - \frac{k}{nm} \sum_{j=1}^R (k+j-1)^{-2} \sum_{r=j}^R n_r \right] - \frac{\psi}{k} [1 - \phi(\varpi^k)]. \quad (17)$$

These values provide a second approximation to the root, which can then be found by a process of successive linear interpolation and extrapolation. Final adjustments may be made, if required, by the usual maximum-likelihood iterative procedure: this will usually be unnecessary, but the quantities (14) can be calculated, to provide an estimate of the asymptotic variance-covariance matrix of the estimates.

As with the moment method, extreme samples can occur for which the equations (15) and (16) have no solutions with $k > 0$. Unfortunately, the function $\psi(k)$ is of considerably more complicated a form than the function $\zeta(\varpi)$ occurring in the moment method, and the existence problem is correspondingly more difficult of solution. It can be shown that

$$\begin{aligned} \psi(0) &= 1, \\ \psi'(0) &= \frac{1}{n} \sum_{j=2}^R \frac{1}{(j-1)} \sum_{r=j}^R n_r + \frac{1}{2} \log \alpha, \end{aligned}$$

where

$$\phi(\alpha) = 1/m,$$

and that

$$\lim_{k \rightarrow \infty} \psi(k) = m(1 - e^{-\theta})/\theta,$$

where

$$\theta = \frac{2}{nm} \sum_{j=2}^R (j-1) \sum_{r=j}^R n_r.$$

Clearly the conditions

$$\psi'(0) > 0, \quad \lim_{k \rightarrow \infty} \psi(k) < 1$$

are sufficient to ensure the existence of at least one solution, and I would conjecture that they are necessary, and also sufficient to ensure uniqueness, but have been unable to prove these results.

The question remains: What action is to be taken when the maximum-likelihood (or moment) method fails to give an acceptable solution? It seems reasonable to hope that one or other of the two limiting forms of the negative binomial, the Poisson distribution and Fisher's 'logarithmic series' distribution, will provide an adequate fit for all but some pathologically extreme (and highly improbable) samples which are unlikely to be fitted satisfactorily by any meaningful distribution.

9. AN EXAMPLE OF THE MAXIMUM-LIKELIHOOD CALCULATIONS

The data are those already used in § 4; details of the calculations are shown in Table 3. The quantity Σn_r , used in the calculation of ϕ , is tabulated in the third column of Table 3 (b), and the remainder of the calculations are shown in some detail in Table 3 (a). In fact, this table shows far more detail than need be recorded in practice; if the calculations are made on a desk machine only the first three columns of Table 3 (b) and the successive values of k , ϖ and ψ need be written down. (The weighted sums of quotients in the second column of Table 3 (a) can be accumulated on the machine, and they, together with the values of ϕ , $\log \varpi$ and ϖ^k , are used as soon as they are calculated, and need not be written down.) However, ϕ and ϖ^k should be recorded for the first trial value, for use in evaluating ψ' , and if final adjustments are to be made by the iterative process, all the quantities tabulated in Table 3 (a) should be recorded for the last trial value of k . The fourth and fifth columns of Table 3 (b) are recorded for the purpose of calculating $\Sigma(k+j-1)^{-2} \sum_{r=j}^R n_r$; the fourth, $(0.63+j-1)^2$, for use in calculating ψ' ; the fifth, $(0.493+j-1)^2$, for use in calculating the variance-covariance matrix. The only remaining values required in this section are

$$nm = 110, \quad m = 3.4375.$$

As an initial trial value the moment estimate of k , 0.633, was rounded to two decimal places, giving

$$\psi(0.63) = 0.9922.$$

As a guide to the location of the second trial value, ψ' was evaluated for $k = 0.63$. Accumulating on the machine,

$$\sum_{j=1}^R (0.63+j-1)^{-2} \sum_{r=j}^R n_r = 92.2942,$$

whence

$$\begin{aligned} \psi' &= \frac{(0.9922 - 0.3179)(0.7008 - 0.5286)}{0.2094} - \frac{0.9922 \times 0.3905}{0.63} \\ &= -0.0605. \end{aligned}$$

The second trial value was therefore taken as

$$0.63 - (1 - 0.9922)/0.0605 = 0.50.$$

The fourth value, $k = 0.493$, gives a value of ψ equal to 1 to four decimal places. There is no hope of improving further on this estimate by the trial-and-error method, without taking more decimal places, and using a more complicated interpolation formula, in the

Table 3. *Maximum-likelihood calculations for the example of § 8*

(a) Trial-and-error calculations

k	$\Sigma(k+j-1)^{-1} \sum_{r=j}^R n_r$	ϕ	ϖ	$\log_e \varpi$	ϖ^k	ψ	ψ'
0.63	77.0905	0.4415	0.2321	-1.46059	0.3984	0.9922	-0.0605
0.50	91.9239	0.4178	0.2124	-1.54930	0.4609	0.99952	
0.491	93.2185	0.4161	0.2110	-1.55590	0.4658	1.00016	
0.493	92.9270	0.4165	0.2113	-1.55450	0.4647	0.99996	

(b) Subsidiary tabulations

j	n_j	$\sum_{r=j}^R n_r$	$(j-0.37)^2$	$(j-0.507)^2$	Expected values of n_j			
					Truncated Poisson	Truncated negative binomial		Log series
						Moments	M.L.	
1	11	32	0.3969	0.243049	4.01	10.31	10.80	13.32
2	6	21	2.6569	2.229049	6.64	6.44	6.36	5.85
3	4	15	6.9169	6.215049	7.33	4.33	4.17	3.43
4	5	11	13.1769	12.201049	6.07	3.01	2.87	2.26
5	0	6	21.4369	20.187049	4.02	2.13	2.03	1.59
6	1	6	31.6969	30.173049	2.22	1.53	1.47	1.16
7	0	5	43.9569	42.159049	1.05	1.11	1.07	0.88
8	2	5	58.2169	56.145049	0.43	0.81	0.79	0.67
9	1	3	74.4769	72.131049	0.16	0.60	0.59	0.53
10	0	2	92.7369	90.117049	0.05	0.44	0.44	0.42
11	1	2	112.9969	110.103049	0.02	1.29	1.41	1.89
12	0	1	135.2569	132.089049				
13	1	1	159.5169	156.075049				

table of ϕ . There is, in fact, no great need of any further improvement, but, if desired, final increments to k and ϖ can be calculated using the variance-covariance matrix, which requires calculation in any case.

The fifth column of Table 3 (b) shows values of

$$(0.493 + j - 1)^2,$$

and from these values (accumulating quotients on the machine)

$$\Sigma (0.493 + j - 1)^{-2} \sum_{r=j}^R n_r = 145.196464173.$$

Then, from this value and the entries in the last line of Table 3 (a), the estimate of the variance-covariance matrix, calculated from formulae (14), is

$$[V] = \begin{bmatrix} 554.4186 & -94.69298 \\ -94.69298 & 19.792976 \end{bmatrix}^{-1} = \begin{bmatrix} 0.00986279 & 0.0471853 \\ 0.0471853 & 0.276265 \end{bmatrix}.$$

Inserting the values $k = 0.493$, $\varpi = 0.2113$ into the left-hand sides of the maximum-likelihood equations (11) and (12), we obtain 0.006 and -0.0003 respectively. Thus the final increments are

$$\begin{bmatrix} \delta\varpi \\ \delta k \end{bmatrix} = [V] \begin{bmatrix} 0.006 \\ -0.0003 \end{bmatrix} = \begin{bmatrix} 0.000045 \\ 0.00020 \end{bmatrix}.$$

Table 4. The function $\phi(x) = \frac{-x \log_e x}{1-x}$

<i>x</i>	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0	0.0465	0.0798	0.1085	0.1341	0.1577	0.1796	0.2002	0.2196	0.2381
0.1	0.2558	0.2728	0.2891	0.3049	0.3201	0.3348	0.3491	0.3629	0.3764	0.3896
0.2	0.4024	0.4149	0.4271	0.4390	0.4507	0.4621	0.4733	0.4843	0.4950	0.5056
0.3	0.5160	0.5262	0.5362	0.5461	0.5558	0.5653	0.5747	0.5839	0.5930	0.6020
0.4	0.6109	0.6196	0.6282	0.6367	0.6451	0.6533	0.6615	0.6695	0.6775	0.6854
0.5	0.6931	0.7008	0.7084	0.7159	0.7233	0.7307	0.7380	0.7451	0.7522	0.7593
0.6	0.7662	0.7731	0.7800	0.7867	0.7934	0.8000	0.8066	0.8131	0.8195	0.8259
0.7	0.8322	0.8385	0.8447	0.8509	0.8570	0.8630	0.8690	0.8750	0.8809	0.8868
0.8	0.8926	0.8983	0.9041	0.9097	0.9154	0.9209	0.9265	0.9320	0.9374	0.9429
0.9	0.9482	0.9536	0.9589	0.9642	0.9694	0.9746	0.9797	0.9848	0.9899	0.9950

The maximum-likelihood estimates are therefore

$$\hat{\varpi} = 0.2113 \pm 0.0993, \quad \hat{k} = 0.493 \pm 0.526.$$

The variance-covariance matrix for the moment estimates, recalculated in terms of the maximum-likelihood estimates to provide a more valid comparison, is

$$\begin{bmatrix} 0.01364125 & 0.0693156 \\ 0.0693156 & 0.405868 \end{bmatrix}.$$

The ratio of the determinants of the two matrices, converted to a percentage efficiency, is

$$100 \times 0.000498/0.000732 = 68.1 \text{ \%}.$$

The efficiency is much the same for the two parameters, the actual values being 72.3 % for ϖ , and 68.1 % for k .

The maximum-likelihood estimate of k does not differ significantly from 0, which suggests that the data might be adequately fitted by Fisher's logarithmic series distribution. This, in fact, proves to be the case. The last four columns of Table 3 (b) show the expected numbers of cells with j breaks on the basis of four fitted distributions, a truncated Poisson (fitted by maximum likelihood), the two negative binomials fitted above, and the logarithmic series distribution (fitted by maximum likelihood). The fit of the Poisson distribution is obviously very poor; equally obviously the other three are all very good. The negative binomials

appear, superficially, to give a slightly better representation of the data, but this impression is largely due to the very good agreement in the first group. In fact the justification for using the negative binomial rather than the logarithmic series distribution comes, not from the data presented here, but from the whole series of experiments, only one of which is used here. In this series some distributions could be fitted adequately by the truncated Poisson, some by the logarithmic series distribution, and some not very satisfactorily by either, but the negative binomial form, with appropriate k , gave a good fit in all cases.

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